

Optimal Clusters

DANA 4840

```
library(factoextra)

## Loading required package: ggplot2

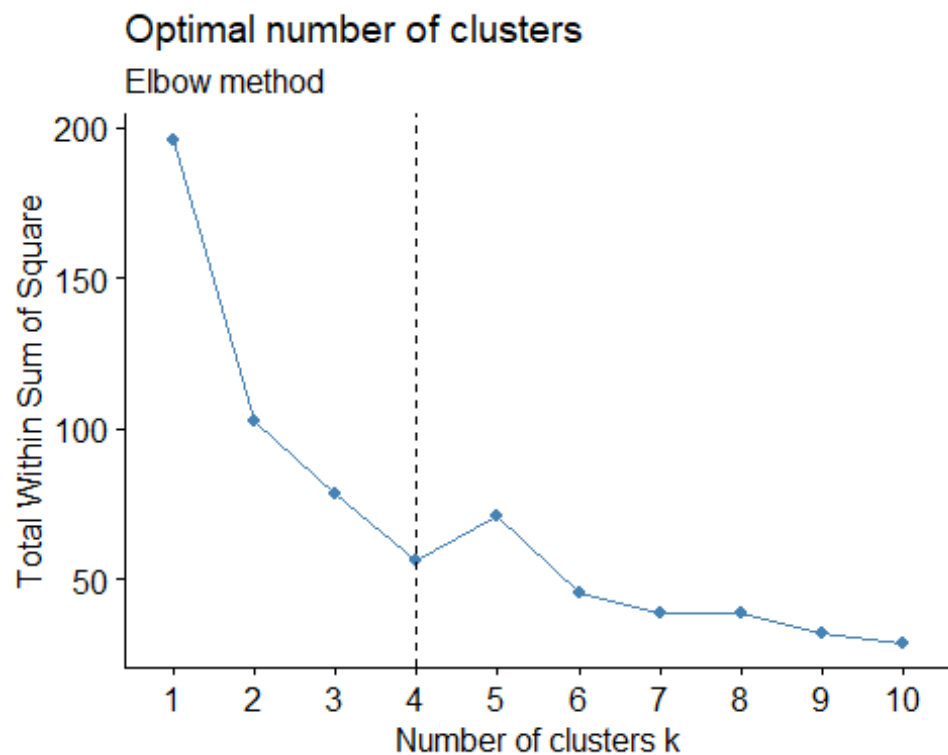
## Welcome! Want to learn more? See two factoextra-related books at
https://goo.gl/ve3WBa

# Standardize the data
df <- scale(USArrests)
head(df)

##           Murder  Assault  UrbanPop      Rape
## Alabama  1.24256408 0.7828393 -0.5209066 -0.003416473
## Alaska   0.50786248 1.1068225 -1.2117642  2.484202941
## Arizona   0.07163341 1.4788032  0.9989801  1.042878388
## Arkansas  0.23234938 0.2308680 -1.0735927 -0.184916602
## California 0.27826823 1.2628144  1.7589234  2.067820292
## Colorado  0.02571456 0.3988593  0.8608085  1.864967207
```

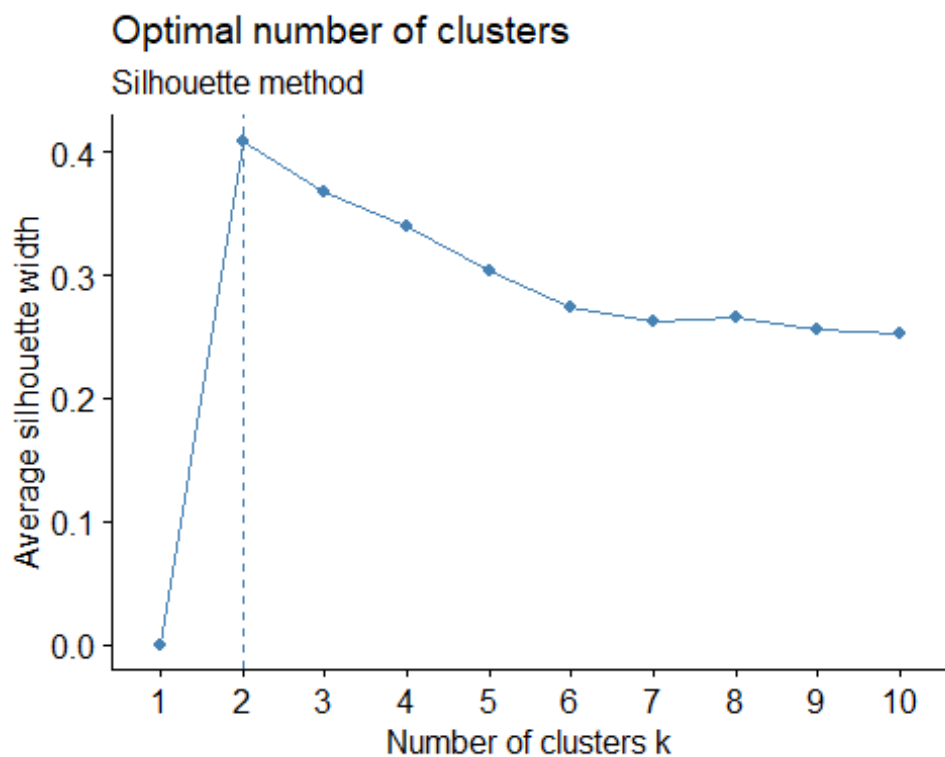
Elbow method

```
fviz_nbclust(df, kmeans, method = "wss") +
  geom_vline(xintercept = 4, linetype = 2) +
  labs(subtitle = "Elbow method")
```



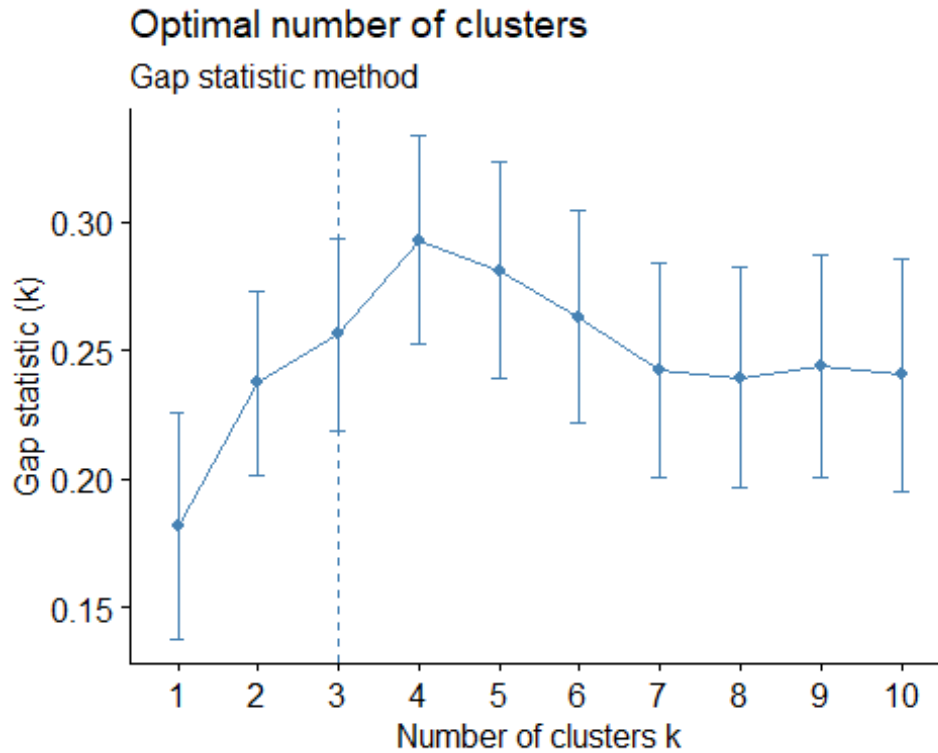
Silhouette method

```
fviz_nbclust(df, kmeans, method = "silhouette")+  
  labs(subtitle = "Silhouette method")
```



Gap statistic

```
# nboot = 50 to keep the function speedy.  
# recommended value: nboot= 500 for your analysis.  
# Use verbose = FALSE to hide computing progression.  
set.seed(123)  
fviz_nbclust(df, kmeans, nstart = 25, method = "gap_stat", nboot = 50)+  
  labs(subtitle = "Gap statistic method")
```



- Elbow method: 4 clusters solution suggested
- Silhouette method: 2 clusters solution suggested
- Gap statistic method: 4 clusters solution suggested

According to these observations, it's possible to define $k = 4$ as the optimal number of clusters in the data.